Blockchain-enabled Federated Learning Data Protection Aggregation Scheme with Differential Privacy and Homomorphic Encryption in IIoT

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***Abstract*—With rapid growth in data volume generated from different industrial devices in IoT, the protection for sensitive and private data in data sharing has become crucial. At present, federated learning for data security has arisen, and it can solve the security concerns on data sharing by model sharing on Internet of mutual distrust. However, the hackers still launch attack aiming at the security vulnerabilities (e.g., model extraction attack and model reverse attack) in federated learning. In this article, to address the above problems, we first design an application model of blockchain-enabled federated learning in Industrial Internet of Things (IIoT), and formulate our data protection aggregation scheme based on the above model. Then, we give the distributed K-means clustering based on differential privacy and homomorphic encryption, and the distributed random forest with differential privacy and the distributed AdaBoost with homomorphic encryption methods, which enable multiple data protection in data sharing and model sharing. Finally, we integrate the methods with blockchain and federated learning, and provide the complete security analysis. Extensive experimental results show that our aggregation scheme and working mechanism have the better performance in the selected indicators.**

***Index Terms*—Blockchain, differential privacy, federated learning, homomorphic encryption, privacy protection.**

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I. Introduction

With the boom of modern Information and Communications Technology (ICT), the fourth industrial revolution (Industry 4.0) has overturned the traditional industrial development model. Industrial Internet of Things (IIoT), where the huge amounts of data are generated by the connected industrial devices every day, is considered as one of the key technologies for the implementation of Industry 4.0. The characteristics of traditional industrial data were small quantity, simple construction, and unidirectional transmission, while, the features of data in IIoT are massive data, complicated structure, and two-way transmission. Business application and direction and path of data flow are more complicated, and data types and protection demand are diverse. It should not be surprising that the safety and security requirements in IIoT are generally stricter than those found in a typical IoT scenario [1]. Therefore, how to protect and use these valuable data in IIoT to share in efficient, secure, and economic ways becomes a burning issue for owners and providers. In recent years, we have seen an unprecedented growth of demand in physical data and logical data security of IIoT.

Federated learning [2] is an emerging machine learning paradigm that enables IIoT in such a way that not only preserves data heterogeneity, availability, and privacy, but also reduces model bias. It provides a distributed machine learning framework with privacy protection, and collaboration in a distributed manner with thousands of participants to iteration training for a particular machine learning model. Since training data are still stored locally in federated learning, this mechanism can not only share the training data of each participant, but also ensure privacy protection for each participant [3]. Compared to traditional machine learning methods, the blockchain-enabled federated learning can enhance the learning efficiency and solve

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the data island problem. Its implementations can be generalized into the three steps: model selection, local model training and aggregation of local models [4]. Federated learning can give significant practical privacy improvements over centralizing all the training data, but there is still no formal guarantee of privacy in this baseline federated learning model [5].

Decentralized storage property of blockchain has paved paths for Industrial Internet and IIoT [6]. Blockchain technology, which emerged after the introduction of an online cryptocurrency named as Bitcoin in 2008 [7], attracted a lot of attentions due to its decentralized, shared, anonymous, trusted, backtracked, and tamper-resistant data structure [8]. Blockchain has the unlimited potential to IIoT applications due to a series of excellent properties. For example, preserving the data users privacy and protecting industrial application data are used in blockchain-based IIoT systems by encryption and authentication strategies.

A. Related Work

The sensitive information security and privacy data protection in IIoT has attracted great interest from academia in recent years. Some hot researches are as follows.

In 2016, Hossain and Muhammad [9] presented a HealthIIoT-enabled monitoring framework, where electro-cardiogram and other health care data were collected by mobile devices and sensors and securely sent to the cloud for seamless access by health care specialists. Signal enhancement, watermarking, and other related analytics have been used to avoid identity theft or clinical error by health care professionals. In 2017, Geyer *et al.* [10] proposed a novel algorithm for client sided differential privacy preserving federated optimization. The method is to hide clients' contributions in the training process. In 2018, in order to provide secure communication in IIoT, Li *et al.* [11] gave a user authentication protocol with privacy protection for IIoT. The security of protocol is proved under a random oracle model, and other security analyses shown that this protocol is robust to resist various attacks. In 2019, Si *et al.* [12], based on blockchain, raised a lightweight IoT information sharing security framework which used the double-chain model composed of data blockchain and transaction blockchain, and strengthened privacy protection by the improved algorithms based on partial blind signature methods.

Since 2020, the research works on blockchain-based privacy protection have been achieving greater development. In [13], a blockchain-based Internet edge model is built for a scalable and controllable IIoT system, and it can take advantages of edge computing and blockchain to establish a privacy preserving scheme which considers other constraints, such as energy cost. In [14], a blockchain empowered secure data sharing architecture for distributed multiple parties is designed, and the data sharing problem is formulated into a machine-learning problem by incorporating privacy-preserved federated learning. The data privacy is well-maintained by sharing the data model instead of revealing the actual data. Meanwhile, the blockchain-based federated learning began to apply to communication device [15] and 5G network [16].

Although the abovementioned researchers have achieved the updated developments and research fruits, some distributed security vulnerabilities (e.g., model extraction attack and model reverse attack) in federated learning need to be fixed. To the best of our knowledge, there are some innovations to solve the problem of distributed security vulnerabilities by providing the security model sharing based on blockchain and federated learning with multiple protection, which are analyzed in subsequent discussion. Our research has great application value in data security of manufacturing supply chain, real-time parameter protection of producing process, and monitoring data security of production equipment from IIoT area.

B. Contributions

Compared with the existing research works, our scheme and working mechanism have great advantages in following aspects.

1) We design an application model of blockchain-enabled federated learning in IIoT, and formulate our data protection aggregation scheme in IIoT in order to solve the security problem in data sharing and model sharing.
2) We propose the distributed K-means clustering based on differential privacy and homomorphic encryption, the distributed random forest with differential privacy and the distributed AdaBoost with homomorphic encryption. All the three methods enable multiple data protection in intricate IIoT scenario.
3) We present a data protection aggregation scheme based on the above methods, and provide the complete security analysis. Extensive experimental results confirm that our scheme and working mechanism have better performance in the selected indicators (such as accuracy and F1-Score).

C. Organization

The rest of this article is organized as follows. Section II demonstrates our application model design and scheme formulation. Section III describes blockchain-enabled federated learning data protection aggregation tactics. What is more, it gives distributed K-means clustering based on differential privacy and homomorphic encryption, the distributed random forest with differential privacy and distributed AdaBoost with homomorphic encryption respectively as well. Section IV discusses the experimental details, and provides the performance evaluation and contrastive analysis. Finally, Section V concludes this article.

II. Application Model Design and Scheme Formulation

A. Application Model Design

In this article, we consider a common distributed federated learning in IIoT scenario with blockchain involved. We try to build a general model which integrates federated learning and distributed architecture of blockchain to be applied well in IIoT scenario. Here, there are four layers in IIoT scene, i.e., service pattern layer, data application layer, communication connection, and perceiving control. The service pattern

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layer containing some industrial applications, such as intelligent transportation, intelligent logistics, intelligent manufacturing, and wisdom medical, etc., relies on the common middleware technology which provides functionalities to seamlessly integrate services and data applications [17]. In data application layer, the cloud computing platform covers both information collaboration platform and service support platform. Here, the former refers to IoT information center, while the latter refers to service infrastructure. In addition, the miners in our model are distributed in the edge computing node in different IIoT application scenarios, where they mine and produce multimodal data. The 4G/5G net and Internet or private network are the communication connection. In IIoT, data-collecting, data-aggregating, and data-transmitting in sensor networks contain private and sensitive information [18]. The top-level perceiving control with tags or sensors is able to automatically sense and acquire important personal information among different IoT terminals.

The application model can be processed in the following steps. First, from data application layer, the data samples are aligned and standardized. Next, the Model A is trained by federated learning, and the trained data are placed in the Block A. The models from B to N operate in a similar vein. The private data are not exchangeable. Then, the N blocks form a chain to blockchain model integration. Finally, the blockchain-empowered federated learning is achieved, and the obtained results are sent to the communication network.

B. Our Proposed Scheme

The distributed federated learning proposed in [2] has emerged as an improved solution to address the shortcomings that the efficiency and accuracy of models in traditional machine learning depend on computing power and training dataset of the centralized server. The federated learning is classified into horizontally federated learning, vertically federated learning, and federated transfer learning based on the fact that how data are distributed among various parties in the feature and sample identity document (ID) space [19]. Horizontal federated learning is introduced in the scenes in which datasets can share the same feature space rather than the different spaces in samples. The horizontal federated learning is summarized as follows:

$$X_i = X_j, Y_i = Y_j, I_i \neq I_j, \forall D_i, D_j, i \neq j \tag{1}$$

where X_i, Y_i, and I_i are used to denote the ith feature space, the ith label space, and the ith sample ID space respectively. The matrix D_i denotes the data held by each data owner i. The expression of the jth is the same as that of the ith.

In [20], the authors proposed a vertical federated-learning method to train a privacy-preserving logistic regression model, which is applicable to such cases that two datasets can share the same sample ID space but differ in feature space. It is summarized as follows:

$$X_i \neq X_j, Y_i \neq Y_j, I_i = I_j, \forall D_i, D_j, i \neq j \tag{2}$$

The federated transfer learning is applied to the scenarios in which two datasets differ in not only samples but also feature space. It is a commendable extension to these existing federated learning methods, which can be summarized as follows:

$$X_i \neq X_j, Y_i \neq Y_j, I_i \neq I_j, \forall D_i, D_j, i \neq j \tag{3}$$

In different IIoT applications, the three federated learning schemes should be used according to the various requirements.

The data from the different industrial scenarios (e.g., onboard data, control sensor data, and intelligent terminal data, etc.) are trained in local model, and then the local results are uploaded to global aggregation and update the global model. The global results are used as the input of federation model. Meanwhile, the data protection aggregation tactics with differential privacy and homomorphic encryption aiming at K-means, random forest, and AdaBoost are used to the above federation model. Our data protection aggregation scheme is showed in Fig. 1.

Algorithm 1: K-means Clustering Based on Differential Privacy and Homomorphic Encryption.

Input: encrypted dataset $D' = \{x^{(i)} | 1 \leq i \leq n\}$ (privacy budget is $\varepsilon/2^n$; positive integer $g = n + 1$), clustering number k.
Output: center, $\{c | x^{(i)} \epsilon \mathrm{D}', i = 1, \ldots, n; 1 \leq c \leq k\}$.
1: getting centroids $\mu_j, 1 \leq j \leq k$ //Selecting the point farther from the sample as the initial clustering center.
2: **repeat**
3: **for** i=1 to n **do**
4: **for** j=1 to k **do**
5: **for** c=1 to k **do**
6: $c^{(i)} = \arg_j \min \|x^{(i)} - \mu_j\|^2$
7: $\mu_j = \frac{\sum_{i=1}^{n} x^{(i)}|_{c^{(i)}=j}}{sum_{i=1}^{n} 1|_{c^{(i)}=j}}$
8: **end for**
9: **end for**
10: **end for**
11: **until** //Distortion function convergence
12: $J(c, \mu) = \sum_{i=1}^{m} \|x^{(i)} - \mu_{c^{(i)}}\|^2$
13: **return** center

III. Blockchain-Enabled Federated Learning Data Protection Aggregation Tactics

Because of access restrictions for enterprise data and privacy concerns for individual data, we design the novel blockchain-enabled federated data protection aggregation tactics learned over the Decentralized Autonomous Organizations (DAOs) instead of the data island. Our data protection strategy covers valuable information of businesses and fine-grained private data of users.

A. Distributed K-Means Clustering Based on Differential Privacy and Homomorphic Encryption

The data object in dataset needs to be treated by 0–1 standardization. It maps data from various dimensions to data in [0, 1]. The mapping function is as follows:

$$\overline{x} = \frac{x - \min_i}{\max_i - \min_i}, i = 1, 2, \ldots, k \tag{4}$$

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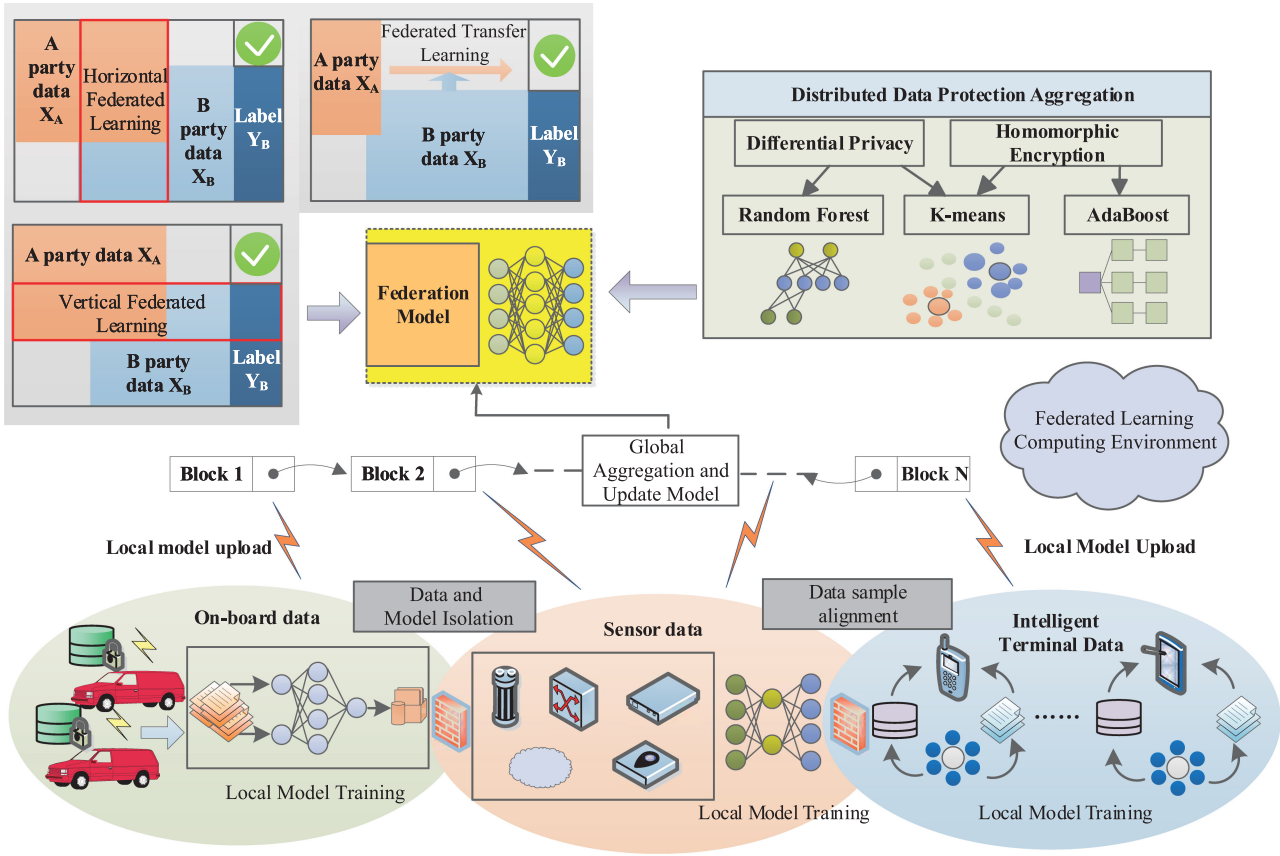


Fig. 1. Our data protection aggregation scheme.

where min_i is the minimum value and max_i is the maximum value in the ith data.

In the dataset after normalization, the maximum is "1" and the minimum is "0" for an arbitrary dimensionality in data object. The data normalization processing before executing clustering algorithm helps to promote the convergence speed and accuracy.

It is the way that randomly picking k data points for the initial cluster center of traditional K-means clustering algorithms can lead to a large error in the clustering results. If the Laplace noise is added in iteration in order to achieve differential privacy protection, the obtained results may diverge even further from the real outcome. Here, we improve the selection method of the initial cluster center, i.e., k points of $(k+1)$ division $(m/(k+1), m = 1, 2, \ldots, k)$ are used as the initial cluster center.

Both the datasets D_0 and D_1 containing d attributes are supposed as the neighboring datasets. When K-means clustering is performed subsequent to normalization of the attributes in D_0 and D_1, the central points in any cluster are the quotient of the sum sum_j of data object and the number num_j of data object in this cluster, i.e. sum_j/num_j. As for num_j, if one record at most is assigned to the different cluster in the clustering results of D_0 and D_1, then the sensitivity is $\Delta f = 1$ for the function to query the data object number in a certain cluster. As for sum_j, the maximum value of each attribute is "1" because of data normalization processing, which leads to the fact that the sensitivity is $\Delta f \leq d$ for the function to calculate the sum of data objects in a cluster. To sum up, the sensitivity is $\Delta f \leq d + 1$ in entire query sequence for our method.

Meanwhile, the value for privacy budget ε needs to be continually adjusted in iteration due to the unknown iteration number. Experience indicates that the effect of early iteration is greater than that of later iteration on clustering results. Therefore, the privacy budget allocation method that we used is that the assigned privacy budget is $\varepsilon/2^n$ while the nth iteration is updated.

In addition, the elbow method is used to compute the optimal k value in this article, to achieve the best clustering effect. The core indicator of elbow method is SSE (the sum of squares due to error), which is the clustering error of all samples and it represents the clustering effect:

$$SSE = \sum_{i=1}^{k} \sum_{p \in \mu_i} |p - m_i|^2 \tag{5}$$

where μ_i is the ith cluster. p is the sample point in μ_i. m_i is the center of mass of μ_i (i.e., the mean of all samples).

In order to strengthen data protection capability, we add the homomorphic encryption. The homomorphic encryption can address these concerns as it allows any third party to operate on the encrypted data without decrypting it in advance [21], for example, Paillier Cryptosystem, processing additive homomorphism, is a representative homomorphic encryption scheme.

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Algorithm 2: The Improved Differential Privacy Protection Method in Random Forest.

Input: dataset D, privacy budget ε, sensitivity, the number of generated decision tree n_{trees}, the maximum depth of each tree $\max_depth + 1$, the number of candidate feature $n_{features}$.

Output: Accuracy.

1: The data in dataset are distributed to the individual nodes by Bagging.
2: Each node selects $n_{features}$ in parallel to form a set F.
3:
$$\varepsilon' = \frac{\varepsilon}{n_{trees}}$$
4:
$$\varepsilon'' = \frac{\varepsilon'}{\max_ depth + 1}$$
5: **for** i=1 to $n_{features}$ **do**
6: leftsize+=noisyCount(sensitivity, ε'')
7: leftreal+=noisyCount(sensitivity, ε'')
8: gini $_1$ = leftreal / leftsize *(1 -leftreal/leftsize)
9: rightsize+=noisyCount(sensitivity, $\varepsilon''/n_{\text{features}}$)
10: rightreal+=noisyCount(sensitivety, $\varepsilon''/n_{\text{features}}$)
11: gini $_2$ = rightreal / rightsize *(1-rightreal/ rightsize)
12: gini $= \max(\text{gini}_1, \text{gini}_2)$
13: **end for**
14: Selecting the optimal segmentation features by min($gini$).
15: Building CERT decision tree, and if the node is a leaf node, the function of counting value is to add the moderate noises. Next, the type is determinate.
16: Inputting testing dataset.
17: $best_{\text{feature}}$ +=noisyCount(sensitivity, ε'')
18: Repeating 3 to 16 until get n_{trees}.
19: Accumulator aggregates classification results.
20: Outputting classification results by majority voting.
21: Calculating the Accuracy.

Algorithm 3: The Improved Paillier Homomorphic Encryption Scheme in AdaBoost.

Input: Encrypted dataset $D' = \{x^{(i)} \mid 1 \le i \le n\}$ (privacy budget is $\varepsilon/2^n$; positive integer $g = n + 1$).

Output: every $x^{(i)}$ category c, weak classifier $G_m(x)$.

1: Calculating the sample weights, and setting the initial weight of each sample to 1/n.
2: **repeat**
3: Training a weak classifier $G_m(x)$.
4: Calculating error rate $\varepsilon = \frac{N_{error}}{N_{all}}$.
5: Calculating the weight of the weak classifier.
6: $\alpha_i = \frac{1}{2} \ln \frac{1-\varepsilon_i}{\varepsilon_i}$
7: Updating weight D_{m+1}.
8: **until** $m \ge$ Specified value or $\varepsilon \le$ Specified value
9: Combining the weak classifiers.
10: Getting the final strong classifier $f(x) = \sum_{m=1}^{M} \alpha_m G_m(x)$.
11: **return** $\{c \mid x^{(i)} \in D', i = 1, \ldots, n; 1 \le c \le k\}, G_m(x)$

Due to the Paillier Cryptosystem's lower efficiency of encryption and decryption and excessive computing overhead, the setting $g = n + 1$ [22] is adjusted as an improved pattern of the setting $g = 2$ of integer g in [23]. Algorithm 1 gives the K-means clustering process based on differential privacy and homomorphic encryption.

B. Distributed Random Forest With Differential Privacy and Distributed AdaBoost With Homomorphic Encryption

1) Bagging [24] is a typical parallelization ensemble learning method in machine learning. Every two individual learners are weak correlation in Bagging. Random forest [25] is an extension version of Bagging. The diversity of base learner in random forest comes from not only sample disturbance but also attribute disturbance, both of which make the generalization performance of ensemble be promoted. The promoted generalization performance is directly proportional to the increasing of difference between every two individual learners [8]. Because random forest is based on multiple decision-making trees, we design that the nodes in blockchain are used as the nodes in a decision-making tree and a forest is made up of multiple chains. In this way, the data in training set are guaranteed not to be tampered by the security of blockchain. In addition, differential privacy can preserve valuable or sensitive data by adding the specific value ε of noise to protect individual privacy [26]. It should be noticed that we introduce differential privacy [27] into random forest. Here, the purpose is to protect the data private information when the big datasets are classified, despite of the lesser impact on classification accuracy. First, the given privacy budget ε is distributed evenly to every tree in forest, and it is given as follows:

$$\varepsilon' = \frac{\varepsilon}{n_{trees}}. \tag{6}$$

Because the samples are randomly chosen, some crossover will exist. According to the characteristics of serial composition of differential privacy, the randomly consumed privacy budget is a superposition of the privacy budget consumed in each decision tree. Second, the privacy budget ε' of every tree is evenly distributed to every layer in the tree, i.e.,

$$\varepsilon'' = \frac{\varepsilon'}{\max_ depth + 1}. \tag{7}$$

Because the nodes in every layer are counted and split, the privacy budget ε" is distributed to every node. According to the characteristics of parallel composition for differential privacy, the privacy budget consumed in nodes is not accumulated. If the node is a leaf node, and the function of the counting value is to add the moderate noises by using the privacy budget ε" combined Laplacian mechanism, then the type of leaf node can be determinate; if the node is an intermediate node, and assuming that m features are randomly selected as the candidate features, then the privacy budget is divided into m parts. The role of the abovementioned privacy budget is played in adding the moderate

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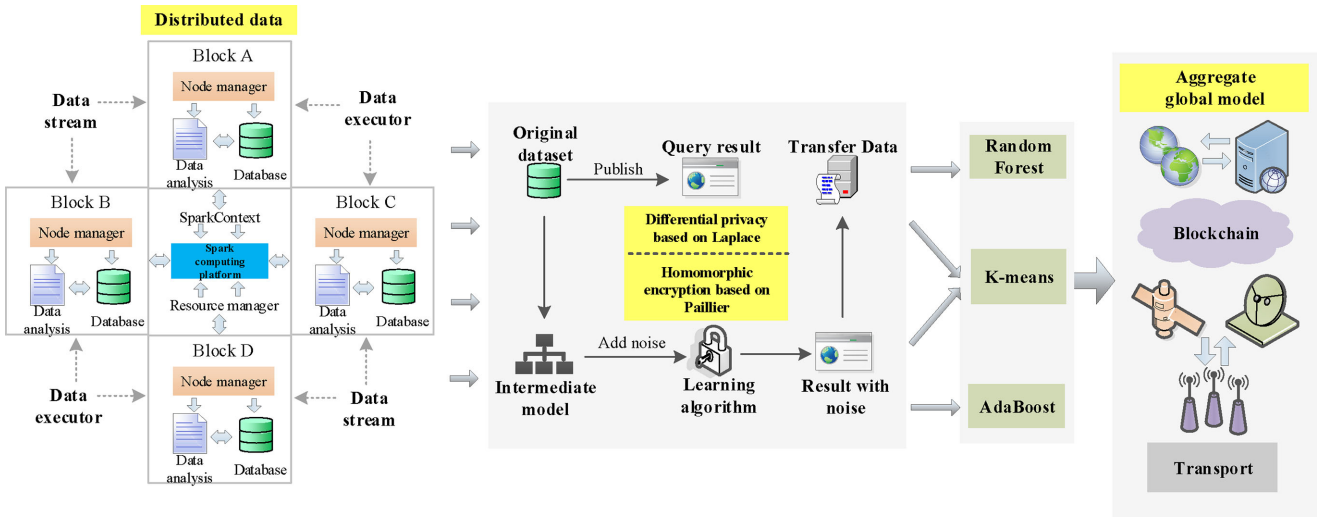


Fig. 2. Working mechanism of our method.

Laplacian random noises in order to achieve the differential privacy protection in the calculation of Gini index. The best split features and the optimal split points are selected. The improved differential privacy protection method in random forest is shown in Algorithm 2.

2) Boosting [28] is a family of algorithms that turn weak learners into strong ones. Adaptive Boosting (AdaBoost) [29] is the best-known example of Boosting family. One point of the algorithm is to keep a distribution or a set of weights on training set. AdaBoost has better results when it runs in a distributed environment, meanwhile it is not prone to overfitting. Here, we introduce the improved Paillier homomorphic encryption scheme in Section III-A into AdaBoost ensemble learning to encrypt distributed original data.

The sample weights need to be initialized as follows:

$$D_1(i) = \frac{1}{n}, i = 1, 2, \ldots, n \tag{8}$$

where D_1 represents the sample weight of the first round of training.

The relation between the weight α_i of weak learner and the weighted classification error itself is shown as follows:

$$\alpha_i = \frac{1}{2} \ln \frac{1 - \varepsilon_i}{\varepsilon_i} \tag{9}$$

where the range of α_i is [0,1].

The formula of the weight update of training set is expressed as follows:

$$W_{i+1}(k) = \frac{W_i(k)}{sum(W_i)} \times \begin{cases} e^{-a_i}, L_i(k) = y_k \\ e^{a_i}, L_i(k) \neq y_k \end{cases} \tag{10}$$

where $W_{i+1}(k)$ is the result that the weight of sample k is updated by calculating weight α_i of weak learner L_i.

The improved Paillier homomorphic encryption scheme in AdaBoost is shown in Algorithm 3.

C. Working Mechanism

The working mechanism of our method is showed in Fig. 2. The RAFT consensus algorithm is used in the proposed scheme. RAFT can achieve the logical separation, and it provides a general manner that the distributed state machine is implemented in a computer system cluster. In this article, each node (such as Block A, B, C, and D) agrees on the same set of state transitions. In addition, we use Spark as our distributed computing platform. First, some distributed data streams are allocated on the multiple nodes by the data executor. In every node, the node manager regulates data analysis and database, and SparkContext is the main entrance to Spark function. Spark computing platform administers all nodes by the resource manager. Next, the privacy of data from multiple nodes is enhanced by the differential privacy and the homomorphic encryption based on Laplace and Paillier Cryptosystem, respectively. The encrypted data are trained with K-means, random forest, and AdaBoost, respectively. Here, K-means clustering uses differential privacy and homomorphic encryption. Random forest uses differential privacy and AdaBoost uses homomorphic encryption. Then, all data results are summarized and the private models are aggregated. They are stored in the cloud. So far, the whole data protection aggregation scheme is finished. Finally, we achieve the goal of the dual protection of data privacy and model privacy by model sharing pattern.

D. Security Analysis

The use of permissioned blockchain establishes a secure mechanism for multiple nodes without mutual trust. We integrate federated learning into the distribution architecture of blockchain with triple protections to address the data security threats.

1) Constructing tamper-resistant blockchain: In a blockchain system, data can only be added and searched (private data and sensitive data cannot be searched), but not be modified and deleted. The tamper-resistant character of blockchain is got by

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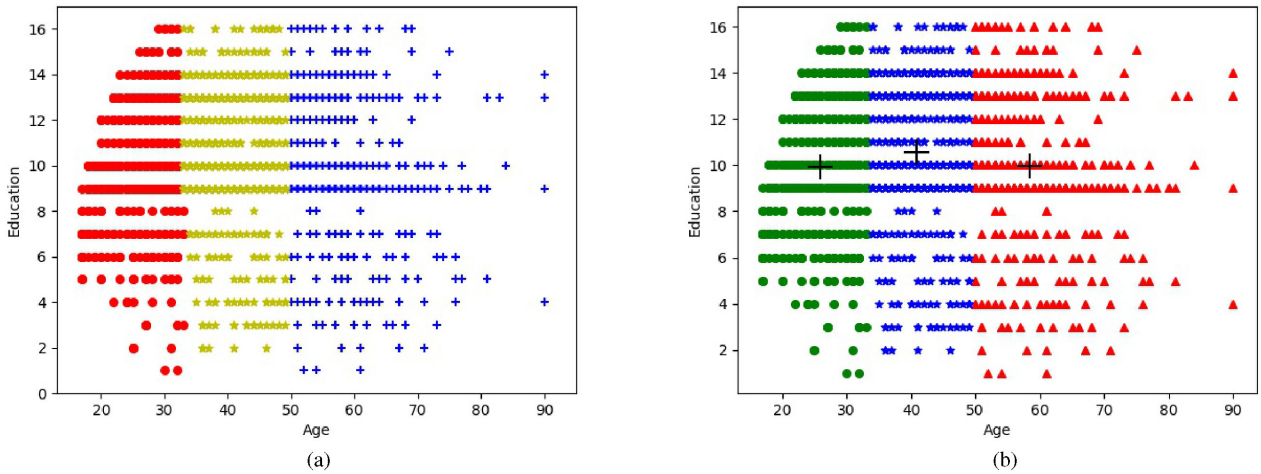


Fig. 3. Comparison for clustering effect between (a) Traditional K-means clustering. (b) Improved K-means clustering.

hash and timestamp [7]. In addition, the tamper-proof log of events and management can achieve access control to data [30]. One significant advantage for our data protection aggregation based on blockchain is that the value Internet trust relationship can be commendably built.

2) Achieving homomorphic encryption: Based on the above analysis, the homomorphic encryption not merely affords better privacy protection [20], but also allows to access the encrypted data on the public blockchain for ciphertext audit at any time. Thanks to the above advantages, the homomorphic encryption can achieve some computing functions among distributed ciphertexts.

3) Implementing differential privacy: According to the definition of differential privacy, if each step of data processing follows the requirement of differential privacy, the final results will satisfy differential privacy [31]. In this way, it demonstrates a property which can be called as the postprocessing invariance/robustness [32]. Differential privacy guarantees that presence or absence of any specific party in a dataset does not affect the query output results in database [26].

IV. Experimental Results and Analysis

In this section, the experimental setting and assessment criteria of the present study will be described and the performances of our data protection aggregation method will also be evaluated and analyzed.

A. Experimental Setting and Assessment Criteria

In this article, the experimental environment uses Spark distributed cluster (Spark2.4.5 + Hadoop2.9.2 + Sbt0.13.9 + Scala 2.11.12), and we set one master node and multiple slave nodes. The master node is responsible for running driver and managing cluster. The slave nodes perform specific tasks. Here, we connect the slave nodes to build a chain. On the above platform, we set up the virtual and artificial blockchain experimental environment and test the performance and security of the blockchain-enabled aggregation scheme and working mechanism. Our algorithmic

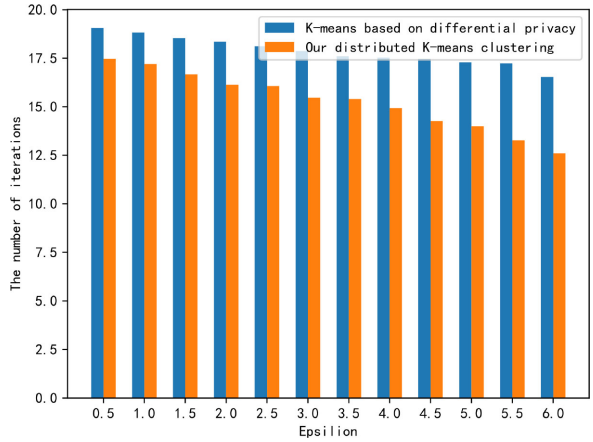


Fig. 4. Comparison for numbers of iteration.

language environment uses Java and Python (Scala for Eclipse + Java + jdk1.8.121 + Python3.7 + PyCharm).

We conduct evaluations of our data protection aggregation scheme on Adult dataset in UCI machine learning dataset library, which is widely used for evaluating some privacy-related machine learning algorithms.

Some assessment criteria to measure our system are indispensable. The two important indicators are used, namely, F1-score and accuracy. The range of F1-score is between 0 and 1. The formulae of the two indicators are defined as follows:

$$\text{F1} - \text{Score} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{11}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{12}$$

where [8]

TP (True positive): It is the number that positive samples are correctly classified as positive samples.

FP (False positive): It is the number that negative samples are incorrectly classified as positive samples.

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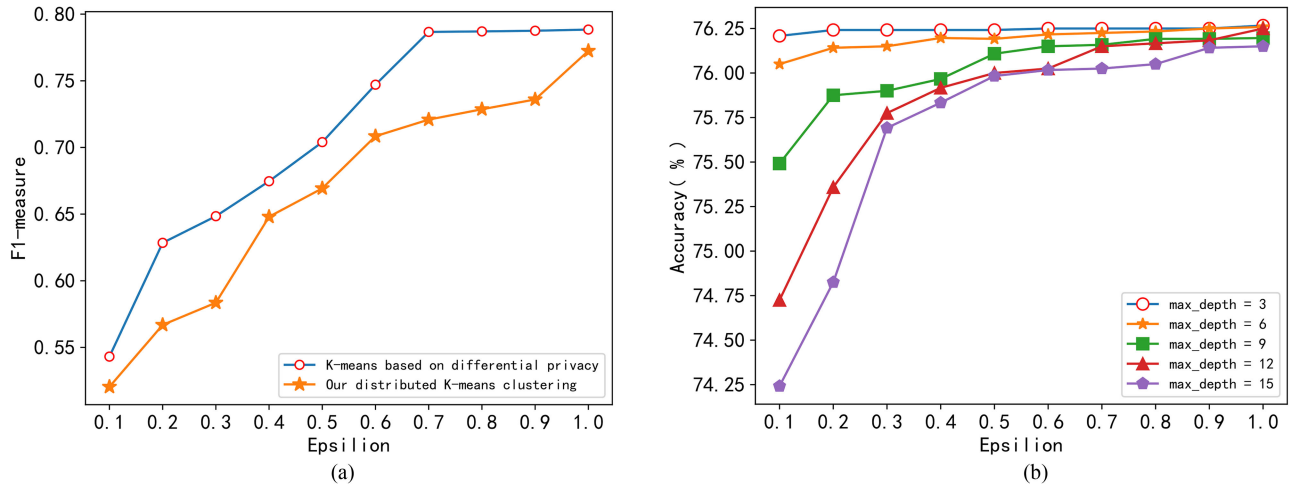


Fig. 5. (a) Comparison for F1-score. (b) Comparison for accuracy.

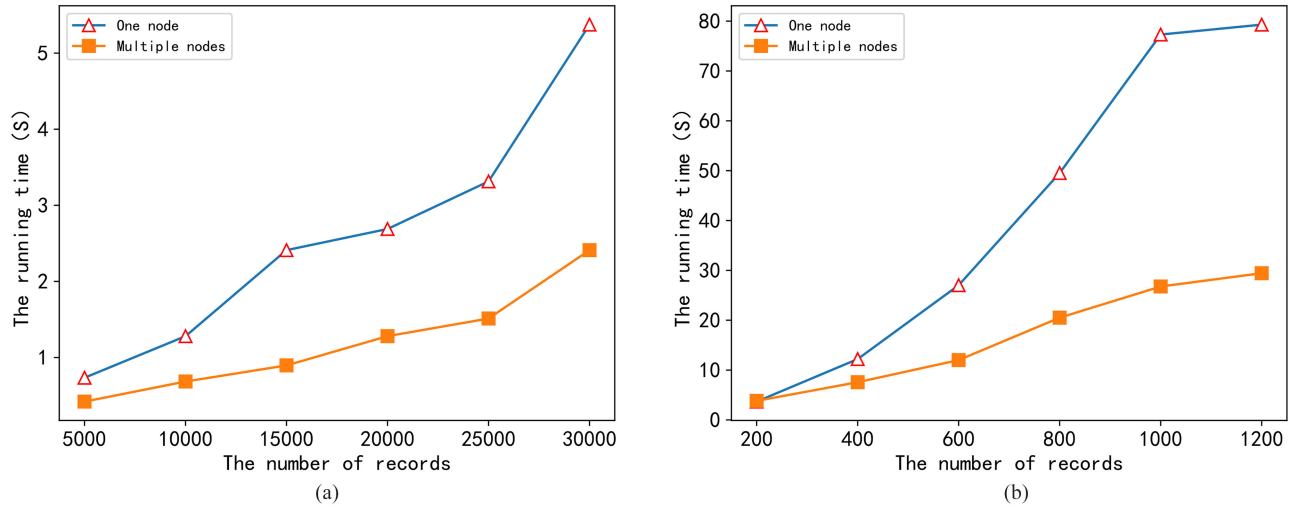


Fig. 6. Change of running time in (a) K-means clustering. (b) Random forest.

TN (True negative): It is the number that negative samples are correctly classified as negative samples.

FN (False negative): It is the number that positive samples are incorrectly classified as negative samples.

B. Performance Evaluation and Contrastive Analysis

First, string data of attribute are converted to numerical data by Python. Then, the numerical data are encrypted, and the ciphertext is trained by federated learning.

According to the elbow method, the optimal number of selected clustering is 3. We select the two important attributes (namely, age and education) in Adult dataset to cluster in order to compare our improved K-means clustering with the traditional one. Their clustering results are shown by the matplotlib of Python in Fig. 3(a) and (b). It is not difficult to find out that our clustering effect is better than the traditional one in Table I. In addition, in order to reduce error, the number of each iteration is the average value taken after the algorithm is called 15 times. The numbers of iteration of two methods are shown in Fig. 4.

TABLE I
CLUSTER CENTER COMPARISON OF TWO METHODS

Traditional clustering	Improved clustering
25.44	9.83
58.48	9.96
40.44	10.57

In Fig. 5(a), the F1-score results of our distributed K-means clustering and the K-means based on differential privacy are exhibited respectively. Fig. 5(b) gives the accuracy with different Laplace noise ε (Epsilion) in the case of different tree depths. It can be easily to discern that our method has the outstanding performance in F1-score and accuracy.

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In Fig. 6(a) and (b), the changes of running time with different data record numbers on one node (centralized system) and multiple nodes (distributed system) of K-means clustering and random forest are exhibited, respectively. We see that our distributed scheme has the shorter running time.

V. Conclusion

In this article, we proposed a blockchain-enabled federated learning data protection aggregation scheme by using differential privacy and homomorphic encryption in IIoT scenario, and gave the related algorithms, working mechanism, and complete security analysis. Compared with traditional or existing works, our methods could solve the security problem in data sharing and model sharing with multiple data protection. Our experiments also showed that our work had the better performance in the selected indicators (such as accuracy and F1-score). What is more, our research will have some positive impacts on improving the level of secure exchange and sharing in industrial data, and the data security support capability in the enterprises of IIoT.

References

[1] K. Tange, M. D. Donno, X. Fafoutis, and N. Dragoni, "A systematic survey of industrial internet of things security: Requirements and fog computing opportunities," *IEEE Commun. Surv. Tut.*, vol. 22, no. 4, pp. 2489–2520, Oct.–Dec. 2020.

[2] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A.y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Mach. Learn. Res.*, 2017 pp. 1273–1282.

[3] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020.

[4] V. Mothukuri, R. M. Parizi, S. Pouriyeh, Y. Huang, A. Dehghantanha, and G. Srivastava, "A survey on security and privacy of federated learning," *Future Gener. Comput. Syst.*, vol. 115, pp. 619–640, 2020.

[5] P. Kairouz *et al.*, "Advances and open problems in federated learning," 2019. [Online]. Available: https://arxiv.org/abs/1912.04977

[6] M. U. Hassan, M. H. Rehmani, and J. J. Chen, "Privacy preservation in blockchain based IoT systems: Integration issues, prospects, challenges, and future research directions," *Future Gener. Comput. Syst.*, vol. 97, pp. 512–529, 2019.

[7] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008. [Online]. Available: http://bitcoin.org/bitcoin.pdf

[8] B. Jia and Y. Liang, "Anti-D chain: A lightweight DDoS attack detection scheme based on heterogeneous ensemble learning in blockchain," *China Commun.*, vol. 17, no. 9, pp. 11–24, 2020.

[9] M. S. Hossain and G. Muhammad, "Cloud-assisted industrial Internet of Things (IIoT) - Enabled framework for health monitoring," *Comput. Netw.*, vol. 101, pp. 192–202, 2016.

[10] R. C. Geyer, T. Klein, and M. Nabi, "Differentially private federated learning: A client level perspective," 2017. [Online]. Available: https://arxiv.org/abs/1712.07557

[11] X. Li, J. W. Niu, M. Z. A. Bhuiyan, F. Wu, M. Karuppiah, and S. Kumari, "A robust ECC-based provable secure authentication protocol with privacy preserving for industrial internet of things," *IEEE Trans. Ind. Informat.*, vol. 14, no. 8, pp. 3599–3609, Aug. 2018.

[12] H. Si, C. Sun, Y. Li, H. Qiao, and L. Shi, "IoT information sharing security mechanism based on blockchain technology," *Future Gener. Comput. Syst.*, vol. 101, pp. 1028–1040, 2019.

[13] K. Gai, Y. Wu, L. Zhu, Z. Zhang, and M. Qiu, "Differential privacy - based blockchain for industrial Internet-of-Things," *IEEE Trans. Ind. Informat.*, vol. 16, no. 6, pp. 4156–4165, Jun. 2020.

[14] Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Blockchain and federated learning for privacy - Preserved data sharing in industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 16, no. 6, pp. 4177–4186, Jun. 2020.

[15] H. Kim, J. Park, M. Bennis, and S. Kim, "Blockchained on - device federated learning," *IEEE Commun. Lett.*, vol. 24, no. 6, pp. 1279–1283, Jun. 2020.

[16] Y. Liu, J. Peng, J. Kang, A. M. Iliyasu, D. Niyato, and A. A. A. El - Latif, "A secure federated learning framework for 5G networks," *IEEE Wireless Commun.*, vol. 27, no. 4, pp. 24–31, Aug. 2020.

[17] L. D. Xu, W. He, and S. Li, "Internet of Things in industries: A survey," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2233–2243, Nov. 2014.

[18] C. Yin, J. Xi, R. Sun, and J. Wang, "Location privacy protection based on differential privacy strategy for big data in industrial Internet-of-Things," *IEEE Trans. Ind. Informat.*, vol. 14, no. 8, pp. 3628–3636, Aug. 2018.

[19] Q. Yang, Y. Liu, T. J. Chen, and Y. X. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019.

[20] R. Nock *et al.*, "Entity resolution and federated learning get a federated resolution," 2018, *arXiv:1803.04035*. [Online]. Available: http://arxiv.org/abs/1803.04035

[21] A. Acar, H. Aksu, A. S. Uluagac, and M. Conti, "A survey on homomorphic encryption schemes: Theory and implementation," *ACM Comput. Surv.*, vol. 51, no. 79, pp. 1–35, 2018.

[22] H. Zhang and C. W. Xu, "Improved probabilistic encryption scheme," *Comput. Eng.*, vol. 36, no. 1, pp. 149–151, 2010.

[23] P. Paillier, "Public - key cryptosystems based on composite degree residuosity classes," *Lecture Notes Comput. Sci.*, vol. 547, no. 1, pp. 223–238, 1999.

[24] L. Breiman, "Bagging predictors," *Mach. Learn.*, vol. 24, no. 2, pp. 123–140, 1996.

[25] L. Rokach, "Ensemble - based classifiers," *Artif. Intell. Rev.*, vol. 33, no. 1, pp. 1–39, 2010.

[26] M. U. Hassan, M. H. Rehmani, and J. Chen, "Differential privacy in blockchain technology: A futuristic approach," *J. Parallel Distrib. Comput.*, vol. 145, pp. 50–74, 2020.

[27] C. Dwork, "Differential privacy," in *Proc. 33rd Int. Conf. Automata, Languages Program.*, 2006, pp. 1–12.

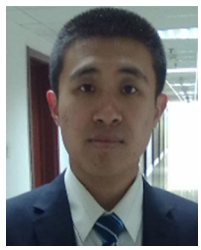
[28] Y. Freund and R. E. Schapire, "A short introduction to boosting," *J. Jpn. Soc. Artif. Intell.*, vol. 14, no. 5, pp. 771–780, 1999.

[29] Y. Freund and R. E. Schapire, "A decision-theoretic generalization of on-line learning and an application to boosting," in *Proc. Eur. Conf. Comput. Learn. Theory*, 1995, pp. 23–37.

[30] W. Viriyasitavat, L. D. Xu, Z. Bi, and D. Hoonsopon, "Blockchain technology for applications in Internet of Things - Mapping from system design perspective," *IEEE Internet Things J.*, vol. 6, no. 5, pp. 8155–8168, Oct. 2019.

[31] C. Dwork, " A firm foundation for private data analysis," *Commun. ACM*, vol. 54, no. 1, pp. 86–95, 2011.

[32] P. C. M. Arachchige, P. Bertok, I. Khalil, D. Liu, S. Camtepe, and M. Atiquzzaman, "Local differential privacy for deep learning," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 5827–5842, Jul. 2020.



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